

Data Plots: Hydrolysis of an Imine

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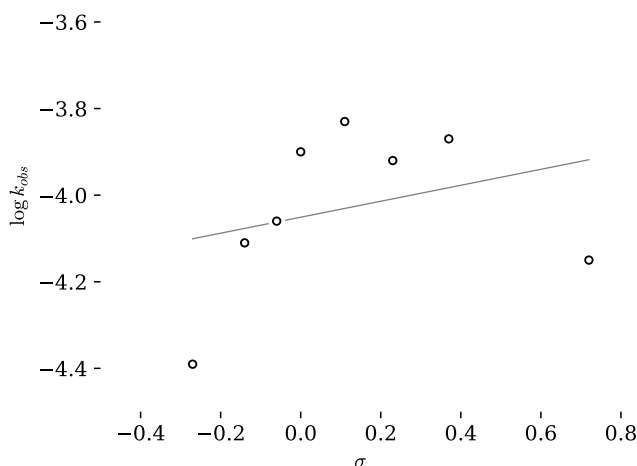
This exploration discusses one of the cases presented in class meeting #14. In the class handout, you were asked to make a Hammett plot and come to a conclusion about the reaction mechanism. This document explores how to interpret the data by plotting and fitting it into mathematical models.¹

Plots: Curved Hammett Plots

This document will present the plots discussed in class. The plots are generated using a *Python* notebook available via Google Colab.² The data comes from an investigation into the mechanism of hydrolysis of salicylaldehyde phenylimines³ This is a fine subject for a student exploration. Someday, I will write one up. There are many rabbit holes to go down here.

The Basic Plot

The following is a Hammett plot of $\log k_{obs}$ vs σ . As you can see, a simple line plot may not be helpful here.



This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

¹ Read chapter 8.4.

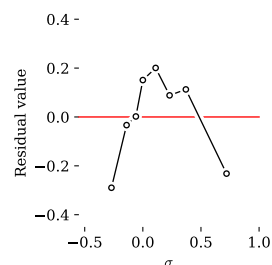
² All plots in this exploration were generated using the interactive *Python* notebook available via Google Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_16/16_Class_Handout_Problems_Imines.ipynb.

³ "Kinetics of hydrolysis of substituted salicylideneanilines." J. Hoffmann, J. Klicnar, V. Štěrba, M. Večeřa, *Coll. Czech. Chem. Comm.*, **1970**, 35, 1387–1398. <http://doi.org/10.1135/cccc19701387>

Figure 1: plot of $\log k_{obs}$ vs. σ . Do you see a bend in the plot? If we fit to a straight line we see that $\rho = 0.18 \pm 0.45$, $R^2 = 0.10$.

←

Figure 2: A plot of the residuals vs. the σ values for the line fit in figure 1. We observe a systematic variation from the model, not random noise. ↓



Two Separate Plots

Below is a Hammett plot of $\log k_{obs}$ vs. σ . As you can see, I divided the points into two groups and performed a line fit on each. Each line is due to a separate Hammett equation.

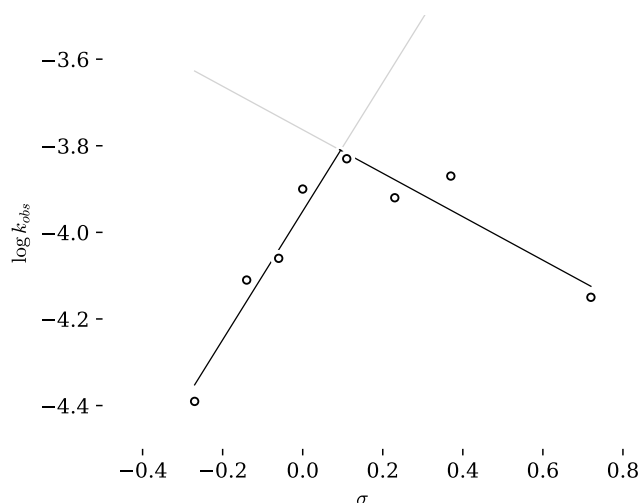


Figure 3: plot of $\log k_{obs}$ vs. σ . I bet you see the bend in the plot now. Of course, I am directing your eye by drawing the two lines.

←

$$y = (1.49 \pm 0.19)x + (-3.95 \pm 0.03)$$

$$R^2 = 0.954$$

$$y = (-0.50 \pm 0.14)x + (-3.76 \pm 0.06)$$

$$R^2 = 0.858$$

That's about it. I have no mathematical model for fitting the two lines where the lesser value controls the rate. I am sure that I could come up with a rate law that would be suitable. I would have to use the Eyring equation, and change the free energy with some function according to the σ values for all the rate constants involved. I haven't gotten around to that yet, and it does seem like a lot of work to implement. In the end, estimating the two slopes is easy with enough data.

Interpretation

So, how do you interpret a Hammett plot where the line is "concave down" and switches from a slope of about $\rho = 1.5$ to a slope of about $\rho = -0.5$? Answer the questions in the class meeting handout, and you will find your way.

Exploration Tips

This exploration has just begun. We have plotted the Hammett plot using data in the paper and found it consistent with the corresponding figure in the paper. My line fits used the top data point twice (five points in the left line, four points in the right line). Was that justified? Feel free to criticize my decision.

Exploration Questions

There is a lot to explore here.

1. **RATE DETERMINING STEP:** The authors lay out a detailed mechanism and provide an argument for how the rds changes. The ρ value for each of the four steps in the proposed mechanism is analyzed. The authors are inconsistent with their symbols, which could also be discussed. It's always fun to pick at small details.
2. **pH-RATE PROFILES:** The authors report a pH-rate profile.³ It is different than typical profiles for imine hydrolysis, likely because of the phenol group of the salicylaldehyde component. It would be fun to make the complete rate law and use it as a model for the pH-rate profile data..
3. **BILL JENCKS:** The authors reference the work of William Jencks, a giant of physical-organic chemistry, who investigated many mechanisms for imines and related groups. You could explore some of this work to highlight how this paper proposes a slightly different mechanistic analysis.
4. **BASE-CATALYZED MECHANISM:** An analysis of the Hammett plot for the base-catalyzed mechanism is presented in the paper. This analysis could be explored as well.
5. **TAFT PARAMETERS:** The authors use Taft's substituent parameters described as σ^0 in this paper, but now usually denoted as σ_I .⁴ They used these values in their LFER investigation for the effect of substituents on the K_a value of the phenol group in the salicylideneanilines. What did the plot reveal? Replot it with the modern sets of σ_m & σ_p values.
6. **TOO MUCH WORK:** I had suggested that I could develop a mathematical model to fit the Hammett plot. There is an opportunity to use the ρ values proposed for all the steps (and perhaps try out various values for the k_{-2} step). But will it work out?

⁴ The values do not match current tables for σ_I values and σ_I does not include the concept of *meta* and *para*. Their use of σ^0 may refer to a superseded data set of substituent parameters reported by Taft. What is the origin of these σ^0 values, and how were they used in their time? Were they widespread in the literature of the 1960s and '70s, or were they a boutique set used by a handful of groups?

As you can see, there was a lot more that could have been done. I could not explore it all in the 10-page body of work I expect in a student exploration report. Pick one, two or three things and dig into them. If your paper is short or the work doesn't have much to go on with, you can choose another paper or explore one of the references to your paper. Start early and give yourself time to swerve around.